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Negative word of mouth for a failed innovation from higher/lower equity brands: Moderating roles of opinion leadership and consumer testimonials

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ABSTRACT

We investigate how higher versus lower-equity brands attract negative word of mouth (NWOM) from consumers when they fail in the marketplace. We find that the dynamics are conditional on the focal consumers' desire for opinion leadership (OL) and the presence/absence of supporting testimonials for the focal brand. For example, among higher-OL consumers, higher-equity brands attract more (less) NWOM than lower-equity brands in the absence (presence) of testimonials. We show that the pattern emerges, in part, because these consumers wish to stand apart from others and be correct in their assessments when they spread NWOM, and such opportunities present themselves best when higher-equity brands, without supporting testimonials, fail in the marketplace. We discuss how the results can help brand managers insulate their brands from consumer backlash in the event of brand failures.

1. Introduction

Introducing new products is vital to business success (Sorescu & Spanjol, 2008; Liao & Cheng, 2014) forcing companies to innovate constantly (Sood & Tellis, 2005). However, the fact is that around 40–90% of all new products fail in the market (Gourville, 2006), and these failures generate negative word-of-mouth (henceforth NWOM) damaging the company's current and future prospects (Richins, 1983). For example, NWOM interrupts the diffusion/adoption process of current customers (Goldenberg, Han, Lehmann, & Hong, 2009), discourages trial/adoption by potential customers (Anderson, 1998; Richins, 1983) and degrades the company/brand's reputation and equity (Bambauer-Sachse & Mangold, 2011).

In this paper, we investigate an important managerial question: If a brand does not live up to its expectations with its innovation (henceforth a failed brand), to what extent does its brand equity (Keller, 1993) discourage or encourage a consumer from spreading NWOM? Extant research is mostly silent on this score as branding and WOM literatures typically follow a silo approach especially when it comes to innovations, and few studies investigate the two in parallel (Lovett, Peres, & Shachar, 2013). This lack of integration is surprising given that if managers wish to plan ahead to mitigate any negative reaction to brand failures, they would need to know if and how WOM spread is conditional on the unique characteristics/equity of their brand.

We conceptualize brand equity as the composite of quality, image, attitude, willingness to pay, and loyalty (Aaker, 1996, 1991; Brady, Cronin, Fox, & Roehm, 2008; Hazée, Van Vaerenbergh, & Armirotto, 2017). We focus on two factors that we argue should moderate the process of WOM spread. The first factor is the focal consumer's degree of opinion leadership (OL) where opinion leadership is the consumer's desire to influence the consumption decisions of others and/or to be a target person that others turn to for consumption advice in a given product category (Flynn, Goldsmith, & Eastman, 1996). The second is the presence/absence of supporting consumer testimonials/endorsements for the focal brand from actual consumers with first-hand satisfactory experiences (Martin, Wentzel, & Tomczak, 2008).

We find that in the absence (presence) of testimonials, higher-equity brands attract more (less) NWOM than lower-equity brands from consumers who show above average desire to be opinion leaders (henceforth, higher-OL consumers, as contrasted to lower-OL consumers). We show that the pattern emerges, in part, because higher-OL consumers wish to stand apart from others and be correct in their assessments when they spread NWOM, and such opportunities present themselves best when higher-equity brands, without any supporting testimonials, fail in the marketplace.

Our research contributes to both theory and practice. For theory, we contribute to three distinct literatures in marketing. First, we contribute to the literature on innovation failures where researchers see higher

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brand equity as an advantage (Brexendorf, Bayus, & Keller, 2015) as well as a disadvantage (Roehm & Brady, 2007) in weathering failure. Our results reconcile these two contradictory assessments – we show that higher equity is an advantage when coupled with consumer testimonials, but a disadvantage when it is not (see Liao & Cheng, 2014 for a similar approach). Second, we contribute to the literatures in branding and Word-of-Mouth (WOM) by showing their interrelationship (Lovett et al., 2013) – failed higher equity brands invite NWOM in the absence of testimonials but thwart NWOM in the presence of testimonials. Finally, our research contributes to the literature on communication strategies in general and consumer testimonials in particular, suggesting that consumer testimonials may discourage NWOM when coupled with higher-equity brands, but encourage NWOM when coupled with lower-equity brands.

For practice, our research informs brand managers how to anticipate and manage NWOM. For example, to minimize consumer backlash in the event of brand failure, managers are better off pairing consumer testimonials with higher-equity brands (accentuating their advantage) instead of pairing them with lower-equity brands (reducing their disadvantage).

We organize the rest of the paper as follows. We start by discussing the theory underlying our hypotheses. Thereafter, we report two studies testing our propositions across two product categories. Finally, we conclude with a discussion of our results and their implications for theory and practice.

2. Theory

2.1. Brand equity and innovation failures

We adopt Keller (1993) definition of brand equity as "the differential effect of brand knowledge on consumer response to the marketing of that brand as compared to if the same product or service did not have that name." Research operationalizes brand equity as the composite of better loyalty, higher quality, higher willingness to pay, positive attitude, and positive image (Brady et al., 2008; Hazée et al., 2017).

Higher-equity brands, compared to their lower-equity counterparts, enjoy numerous advantages such as higher market share (Feinberg, Kahn, & McAlister, 1992), higher perceived quality (Chandon, Wansink, & Laurent, 2000), and better evaluations (Smith, 1993). They are able to charge a price premium (Park & Srinivasan, 1994) and reduce the uncertainty consumers might have about their quality (Erdem, Zhao, & Valenzuela, 2004). Consequently, consumers expect the innovations made by higher-equity brands to be more successful than innovations made by lower-equity brands. They are quicker to adopt such innovations because higher-equity brands, with their reputation, reduce the risk of adopting a new product (Brexendorf et al., 2015).

Yet, about half of all innovations introduced to the market fail (Nejad, Sherrell, & Babakus, 2014), i.e., they do not meet the market's expectations in terms of performance (Liao & Cheng, 2014), and higher equity brands are no exception (Gilbert, 2019). For example, some recent high-profile failures of higher-equity brands include Samsung's Galaxy Note 7 (2016), Amazon's Fire Phone (2014), Google Lively (2008), Microsoft Zune (2005) and McDonald's Arche Deluxe (1996). Since consumers expect more from higher than lower-equity brands (Oliver, 1997), such failures are likely to catch consumers by surprise and generate considerable disappointment (Roehm & Brady, 2007). The fundamental question that we address in this paper is whether a higher-equity brand failure is likely to generate more/less NWOM compared to a lower equity brand failure, and if consumer characteristics such as the desire to be an opinion leader and/or marketing interventions such as consumer testimonials describing first-hand satisfactory experiences with the brand, affect the pattern.

2.2. Negative word-of-mouth and desire to be opinion leaders and stand apart

Word-of-Mouth (WOM) is an informal flow of communication between consumers about their product experiences (Anderson, 1998; East, Hammond, & Wright, 2007) and considered perhaps the most important information source in the innovation diffusion process (Arndt, 1967; Leonard-Barton, 1985; Yang, Hu, Winer, Assael, & Chen, 2012). WOM can be negative, positive, or neutral (Lovett et al., 2013) and a failed performance is likely to generate NWOM about the focal brand (Zhang, Feick, & Mittal, 2014). NWOM is a pervasive and persuasive force on consumers, and highly problematic for marketers (Philp, Pyle, & Ashworth, 2018). It degrades consumer brand judgments (Herr, Kardes, & Kim, 1991) and purchase decisions (Samson, 2006), and, consequently decreases sales (Basuroy, Chatterjee, & Ravid, 2003) and firms' stock prices (Luo, 2009).

Not all consumers are equal in their desire to spread NWOM about a failed brand within a given product category. The literature suggests that higher-OL consumers who are higher in their desire to be opinion leaders are more likely to do so. Specifically, opinion leadership refers to a consumer's desire or tendency to provide consumption-related information to others and/or be a source of such information to others (Childers, 1986). This desire may change across product categories, i.e., consumers may strongly desire to be opinion leaders in some product categories but weakly so in others (e.g., Ruvio & Shoham, 2007; Flynn et al., 1996). For the purpose of our research, and consistent with the literature, we look at consumers ranging from below average to above average in OL with their location on the OL scale defining their degree of OL (Flynn et al., 1996).

Research suggests that, compared to their lower-OL counterparts, higher-OL consumers are more involved with a focal product and have more knowledge in the given product category (Chan & Misra, 1990). They are also more self-centred (Baumgarten, 1975) and like to self-enhance or boost their self-image (Berger, 2014; Zhang et al., 2014) by displaying their expertise to others so that they can stand apart from them (Baumeister, 1998; Chan & Misra, 1990; Wojnicki & Godes, 2012). In sum, they are concerned with building and maintaining their status for shopping related matters. Conversely, lower-OL consumers have neither enough involvement nor desire to be opinion leaders (Richins & Root-Shaffer, 1988) and thus they do not have any reason (or motivation) to build or maintain a status for shopping related matters. Indeed, King and Summers (1970) classify higher and lower-OL consumers as active opinion leaders and passive followers respectively. In the context of our research, this implies that higher-OL consumers will be always looking out for opportunities to stand apart from others and display their expertise, and one way of doing so is to share their experiences about a failed brand (NWOM).

Against this backdrop, let us, first, consider what happens when a higher-equity brand fails in the market. Since consumers do not expect higher-equity brands to fail, and higher equity brands are supposed to have less variance in performance (Keller, 1993), the news of their failure carries a surprise element as well as strong information value. Spreading the news, therefore, will help the higher-OL consumers to stand apart from others in matters of consumption. However, and second, when lower-equity brands fail in the market, the news is neither surprising nor informative since the market does not expect lower equity brands to succeed. Spreading the news, then, would be stating the obvious and not something that would make the higher-OL consumers stand apart from others.

The above discussions suggest a brand-equity by OL interaction linking innovation failures and NWOM. Specifically, an innovation failure from a higher-equity brand should attract *more* NWOM compared to an innovation failure of a lower-equity brand from higher-OL consumers. Notice that we are not saying that lower-OL consumers will not generate NWOM following a brand failure. Indeed, they might be disappointed when a brand fails (see disconfirmation of expectation

paradigm; Oliver, 1997) and spread NWOM. However, since they do not desire to stand apart/seek the status of experts and are comfortable in the role of followers (King & Summers, 1970), whether it is a higher-equity brand or lower-equity brand that fails will not affect their NWOM behaviour as much as it will affect the behaviour of higher-OL consumers. Formally,

H1: Higher-equity brands will attract *more* NWOM compared to lower-equity brands following brand failures from higher-OL consumers.

H2: Finding an opportunity to stand apart from others by providing NWOM about a failed brand will indirectly affect how the brand equity of the failed brand will generate NWOM from higher-OL consumers. Specifically, failed higher-equity brands will provide *more* opportunity to stand apart compared to failed lower-equity brands to higher-OL consumers, which, in turn, will generate more NWOM from them.

2.3. The role of consumer testimonials

Managers introducing innovations are likely to look for ways to anticipate and counter NWOM in the event their innovations fail in the marketplace. Consumer testimonials that are endorsements from actual consumers with a first-hand satisfying experience (Martin et al., 2008) can accomplish this purpose. They can be quite persuasive as long as consumers believe that the customers portrayed in the testimonials represent actual users (Elgaaied-Gambier, Monnot, & Reniou, 2018; Kemp et al., 2017). The question we ask is whether the presence or absence of consumer testimonials will change the NWOM behaviour of higher-OL consumers.

Positive consumer testimonials provide contradictory information in the event of brand failures. The consumer who has experienced a brand failure has had a negative brand experience, whereas the testimonial suggests that another person (featured in the testimonial) has had a positive brand experience. Therefore, contradicting the testimonial by spreading NWOM is somewhat risky for the person doing so because s/he may turn out to be incorrect and thus lose her/his status/credibility as an expert.

Against this backdrop, now consider an innovation failure of a higher-equity brand supported by a consumer testimonial. Since higher-equity brands are perceived highly by the market, consumers believe that they are more likely to provide good performance (Rao, Qu, & Ruekert, 1999) and on a more consistent basis (Keller, 1999, 1993). Accordingly, the consumer testimonial is consistent with the general expectations of higher-equity brands. Therefore, the consumer who has experienced the brand failure may think that her/his own experience is an exception. In that case, spreading NWOM (and thus contradicting the testimonial) may be quite risky since s/he may turn out to be wrong in the end. However, when an innovation from a lower equity brand supported by a consumer testimonial fails in the market, contradicting the testimonial and spreading NWOM is relatively less risky for two reasons. First, lower-equity brands are not supposed to be successful or at least not as much as higher-equity brands (Keller, 1993) such that the testimonial is now an exception rather than the norm because it is inconsistent with the general expectations of the market. Second, even if the consumer turns out to be wrong, she/he can justify her opinion by pointing out the relatively more variability in performance of lower equity brands. The upshot is that failing lower-equity brands provide a better opportunity to spread NWOM if the objective of the consumer is to be correct in her/his assessments and stand apart from others when doing so.

As before, we hypothesize that higher-OL consumers will be more mindful of such risks. Research indicates that these consumers tend to tread carefully when they face contradictory information. This is because higher-OL consumers want to make sure that they are always giving the right advice and not appear as incompetent recommenders (Van Eck, Jager, & Leeflang, 2011; Richins, 1984) and/or risk losing their status in the market (Shi & Wojnicki, 2014). This suggests a brand-equity by OL interaction linking innovation failures and NWOM. However, the pattern of this interaction in the presence of consumer testimonial is reverse of the pattern discussed earlier in the absence of consumer testimonials. Specifically, in the presence of consumer testimonials, an innovation failure from a higher-equity brand should attract *less* NWOM compared an innovation failure of a lower-equity brand from higher-OL consumers. Notice, once again, that we are not saying that lower-OL consumers will not generate NWOM following a brand failure in the presence of consumer testimonials. Since they do not desire to stand apart and, therefore, are less sensitive to the risk of being wrong, whether it is a higher-equity brand or lower-equity brand that fails will not affect their NWOM behaviour as much as it will affect the behaviour of higher-OL consumers. Specifically,

H3: In the presence of consumer testimonials, higher-equity brands will attract *less* NWOM compared to lower-equity brands following brand failures from higher-OL consumers.

H4: Finding an opportunity to stand apart from others by providing NWOM about a failed brand will indirectly affect how brand equity will generate NWOM from consumers. Specifically, in the presence of consumer testimonials, failed higher-equity brands will provide *less* opportunity to stand apart compared to failed lower-equity brands to higher-OL consumers, which, in turn, will generate fewer NWOM from them.

Fig. 1 depicts our conceptual model. Hypotheses 1 and 3 implicate a moderated moderation process (a three-way interaction between brand equity, opinion leadership and consumer testimonials; Fig. 1A). Hypotheses 2 and 4 implicate a moderated mediation process and tests for the mechanism underlying H1 and H3 by investigating the proposed mediation role of the desire among consumers to stand apart when providing NWOM (Fig. 1B).

3. Study 1

Study 1's objective is to test H1 and H3, i.e., to show that NWOM generated by failures among higher and lower-equity brands is conditional on the consumer's desire to be an opinion leader and the presence/absence of consumer testimonials.

3.1. Main study: participants, stimuli, and design

Two hundred seventy-six M-Turk participants (123 females, M_{Age} = 39.38 SD = 12.66, all US residents) took part in Study 1.

In Study 1, the focal innovation is a new facial recognition payment system introduced by a bank to increase convenience for the customers by introducing "cardless" payment. We focus on technology for two reasons. First, research shows that WOM affects diffusion of technological products (Herr et al., 1991) and their adoption requires risk/benefit trade-offs for consumers (Truong, Klink, Simmons, Grinstein, & Palmer, 2017). Second, within the technology spectrum, facial recognition systems that increase consumer convenience are getting traction all over the world (Pascu, 2019).

We varied brand equity (higher and lower equity) using a fictitious brand (Brand A; Chernev & Carpenter, 2001; Biehal & Chakravarti, 1983). Using existing (known) brands would introduce different confounds as they would likely differ in how much experience consumers have had with these brands, how much they trust them, and how they feel about them.

We conceptualize brand equity as the composite of five items – loyalty, quality, willingness to pay, attitude, and image (Aaker, 1996, 1991; Brady et al., 2008; Hazée et al., 2017). In our manipulation, therefore, we describe Brand A as "very positive"/"very high" on all these components, and we describe the lower-equity brand as "about average" on all these components. Appendix A describes our stimuli.

The participants imagined that they were switching to a new bank offering a facial recognition system to facilitate "cardless" payments where the customer simply nods at the device to confirm the payment.

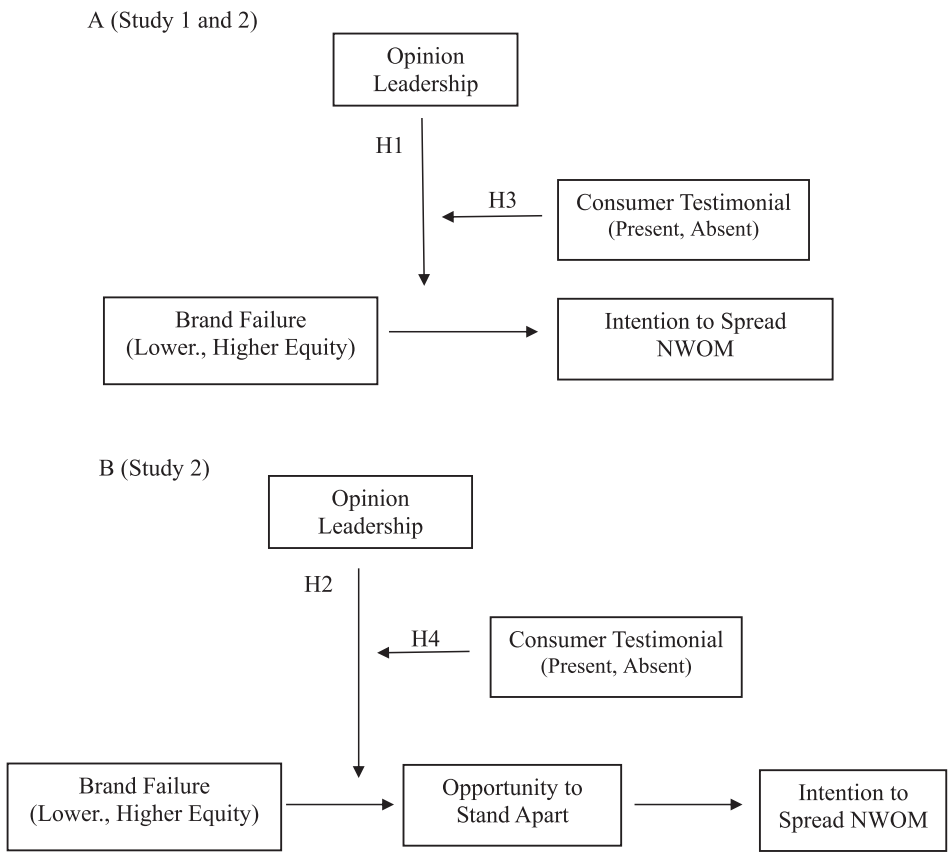

Fig. 1. Process models tested in studies 1 and 2.

Some participants imagined choosing a higher-equity bank offering the tracking device while others imagined choosing a lower-equity bank offering the tracking device. Thereafter, we described the brand failure to all participants. They read that the first time they use the facial recognition payment, the system does not recognize them and they have to resort to paying by credit card.

Thereafter, some participants imagined coming across a positive consumer testimonial, while others did not get any testimonial information. The testimonial described a positive comment from a real customer (Robert Brown): "*I switched to Brand A bank to benefit from the new facial recognition system they offer. This is great! I don't need to carry anything for my payments*". We used the full name of the customer to increase the effectiveness of the testimonial (Martin et al., 2008). We divided the scenario into two parts (pre- and post-purchase) and presented them on different screens (i.e., participants needed to click on "next" to read the post purchase scenario). This was done to mitigate participant fatigue due to a long scenario that may have caused them to pay less attention to the testimonial scenario.

We assigned participants randomly to the four between-subjects experimental conditions obtained by crossing brand equity (higher, lower) and testimonial (present, absent). The sample size, per cell, ranged between 65 and 73 (72 and 73 across higher and lower brand-equity conditions when the testimonial was present, and 65 and 66 when it was absent).

3.2. Measures

Appendix B describes our measures and reports the results of the factor analyses (where applicable) confirming the integrity of treating the items as loading on a single factor.

In Study 1, the independent variable is brand equity (manipulated at two levels, higher and lower), the outcome variable is desire to spread NWOM (continuous variable; three 7-point items; $\alpha = 0.86$), and the two moderators are opinion leadership (continuous variable; six 7-point items, $\alpha = 0.90$) and consumer testimonials (manipulated at two levels; absent, present).

We used three control variables, satisfaction, gender, and age. Since satisfaction is likely to affect the intention to spread NWOM (e.g., East et al., 2007) we measured how satisfied the participants were with the device, with two 7-point items ($\alpha = 0.92$) from Oliver (1980). Males and females have different reasons for spreading NWOM (Zhang et al., 2014). Similarly, older (vs. younger) consumers have different attitudes toward innovations (Lunsford & Burnett, 1992) and likelihood of generating WOM (East, Uncles, & Lomax, 2014).

We checked for our manipulation of brand equity on all five components used in our manipulation (five 7-point items; $\alpha = 0.94$). We checked if participants regarded the facial recognition system as innovative (four 7-point items; $\alpha = 0.90$) and thought it important enough in selecting a bank (one 7-point item). We checked if participants agreed that the innovation had failed by measuring how they scored the device on performance (one 7-point item), to what extent they thought this was a failure of technology (one 7-point item) and to what extent they blamed the bank or themselves for the failure (one 7-point item). Finally, we checked if adding the testimonial affected information load (two 7-point items; $\alpha = 0.72$), scenario comprehension (one 7-point item), argument strength (two 7-point items; $\alpha = 0.80$) and scenario believability (two 7-point items; $\alpha = 0.90$), across our two scenarios.

We presented these items as follows. The participants responded to the outcome variable (NWOM) and thereafter rated the satisfaction and manipulation check items. Thereafter, they completed the opinion leadership scale and the demographic questions. The exception was the brand equity manipulation check items and we detail that below.

Table 1
Study 1: (moderated) moderation model.

A. Dependent Variable: NWOM

	Coefficient	se	t	p	LLCI	ULCI
Constant	4.0455	0.0906	44.6610	0.0000	3.8671	4.2238
Equity (Higher = 1, Lower = 0)	−0.2116	0.1812	−1.1678	0.2440	−0.5685	0.1452
Opinion Leadership (OL)	0.1264	0.0704	1.7937	0.0740	−0.0124	0.2651
Equity × OL	0.2772	0.1410	1.9658	0.0504	−0.0005	0.5549
Testimonial (Present = 1, Absent = 0)	0.0283	0.1819	0.1557	0.8764	−0.3298	0.3865
Equity × Testimonial	−1.0229	0.3639	−2.8111	0.0053	−1.7395	−0.3063
OL × Testimonial	−0.0354	0.1417	−0.2499	0.8028	−0.3145	0.2436
Equity × OL × Testimonial	−0.7778	0.2837	−2.7420	0.0065	−1.3363	−0.2192

B. Conditional effect of brand equity on NWOM on different values of OL and testimonials

OL	Testimonials	Equity coefficient	Se	t	p	LLCI	ULCI
−1 SD Below Mean	Absent	−0.5587	0.3625	−1.5409	0.1246	−1.2726	0.1553
−1 SD Below Mean	Present	−0.5782	0.3579	−1.6157	0.1074	−1.2830	0.1265
+1 SD Above Mean	Absent	1.2394	0.4023	3.0811	0.0023	0.4472	2.0315
+1 SD Above Mean	Present	−0.7869	0.3336	−2.3586	0.0191	−1.4438	−0.1299

NWOM = Negative Word of Mouth, OL = Opinion Leadership, se = Standard Error, LLCI = Lower-Level Confidence Interval, ULCI = Upper-Level Confidence Interval.

3.3. Results

3.3.1. Manipulation checks

We varied the presentation of the brand-equity manipulation checks. Checking for brand equity before the dependent variable might lead participants to guess the purpose of the study, while checking after the dependent variable may contaminate our checks, especially when the brands fail. In order to prevent the order problem, half of the participants answered the manipulation check questions before addressing the dependent variable (but before they read about the failure), and the other half responded to the questions after addressing the dependent variable (and after reading about the failure).

We conducted an ANOVA on the composite manipulation check ($\alpha = 0.94$), with brand equity (higher, lower), testimonials (present, absent), and order (before/after dependent variable) as predictors. As expected, participants rated Brand A's equity significantly higher in the higher-equity conditions compared to the lower-equity conditions (M's of 5.34 and 3.84, $F(1, 270) = 149.97$, $p < 0.0001$) that was unqualified across the testimonial and order conditions (all interaction p's > 0.05).

3.3.2. Attribute innovativeness and importance

We conducted separate ANOVA's to check how innovative/important participants thought the facial recognition device to be, with brand equity and testimonials as predictors. We found no significant main or interaction effects (all p's > 0.05). The innovativeness and importance means for both higher and lower equity brands were significantly above 4.0 or the scale's midpoint (higher equity brand: $M = 5.46$, $t(136) = 13.88$, $p < 0.0001$ and $M = 5.23$, $t(135) = 9.16$, $p < 0.0001$; lower equity brand: $M = 5.4$, $t(137) = 12.38$, $p < 0.0001$ and $M = 4.99$, $t(138) = 7.36$, $p < 0.0001$).

We conducted separate ANOVA's to check how participants rated the performance of the recognition device, to what extent they thought that the technology had failed and to what extent they blamed themselves or the device for the failure, with brand equity and testimonials as predictors. We found no significant main or interaction effects (all p's > 0.05). The performance means for the higher and lower-equity brands were all significantly below 4.0 or the scale's mid-point for performance ($M = 2.80$, $t(136) = -10.31$, $p < 0.0001$ and $M = 2.54$, $t(138) = -11.44$, $p < 0.0001$). The failure means for the two brands were all significantly above 4.0 or the scale's mid-point ($M = 5.53$, $t(136) = 13.17$, $p < 0.0001$ and $M = 5.80$, $t(138) = 14.42$, $p < 0.0001$). Finally, the means for blaming the bank/device were significantly higher than 4.0 or the scale's mid-point for the higher and lower-equity brands, with higher numbers indicating more blame to the device and less blame to the self ($M = 5.54$, $t(136) = 13.04$, $p < 0.0001$ and $M = 5.81$, $t(137) = 15.78$, $p < 0.0001$).

Finally, we compared if adding the testimonial descriptions changes information load, scenario comprehension, scenario believability and argument strength across the different experimental conditions. ANOVAs on these measures with brand equity and testimonials as predictor did not return any significant main or interaction effects (all p's > 0.05).

3.3.3. Hypothesis tests

The tests for H1 and H3 implicate a moderated moderation model (Fig. 1). Specifically, OL (continuous variable; mean centred) moderates how much NWOM (continuous variable) higher/lower equity failing brands (higher/lower equal to 1/0) attract, and the presence or absence of consumer testimonials (present/absent equal to 1/0) further moderates the moderation (see Hayes, 2018). We ran the model with and without the covariates (satisfaction, gender, age) and measurement order for brand equity (presented before/after measuring NWOM). With covariates in place, we found that only satisfaction was significant ($\beta = -0.33$, $t = -4.86$, $p < 0.0001$) with satisfaction reducing NWOM intent. The main results, however, were unchanged with respect to statistical significance at $p < 0.05$ with or without covariates. Table 1 shows the results for the unadjusted model.

Table 1 shows that the coefficient of equity, conditional on OL = zero (sample average) and testimonial = zero (or absent) is insignificant ($\beta = -0.21$, $t = -1.17$, $p = 0.24$) indicating that, in the absence of testimonials and at average OL, both higher and lower-equity failed brands attract the same amount of NWOM. However, the equity by OL interaction is positive and significant ($\beta = 0.28$, $t = 1.97$, $p = 0.05$). The positive sign is consistent with H1 and indicates that, at above-average OL (higher OL-consumers), higher-equity failed brands attract significantly more NWOM than lower equity failed brands (see Fig. 2A). To explicate the interaction, we conduct a spotlight analysis examining the slopes at +1 SD above mean and −1 SD below mean OL. At + 1 SD above mean for OL, higher-equity brands attract significantly more NWOM than lower equity brands (M's of 4.83 and 3.59; $\beta = 1.24$, $t = 3.08$, $p < 0.01$) but there is no such difference at −1 SD below mean OL (M's of 3.57 and 4.13; $\beta = -0.56$, $t = -1.54$, $p = 0.13$).

The presence/absence of testimonials further moderates the equity by OL interaction (three-way-interaction $\beta = -0.78$, $t = -2.74$, $p < 0.01$). The negative sign is consistent with H3. It indicates that, at

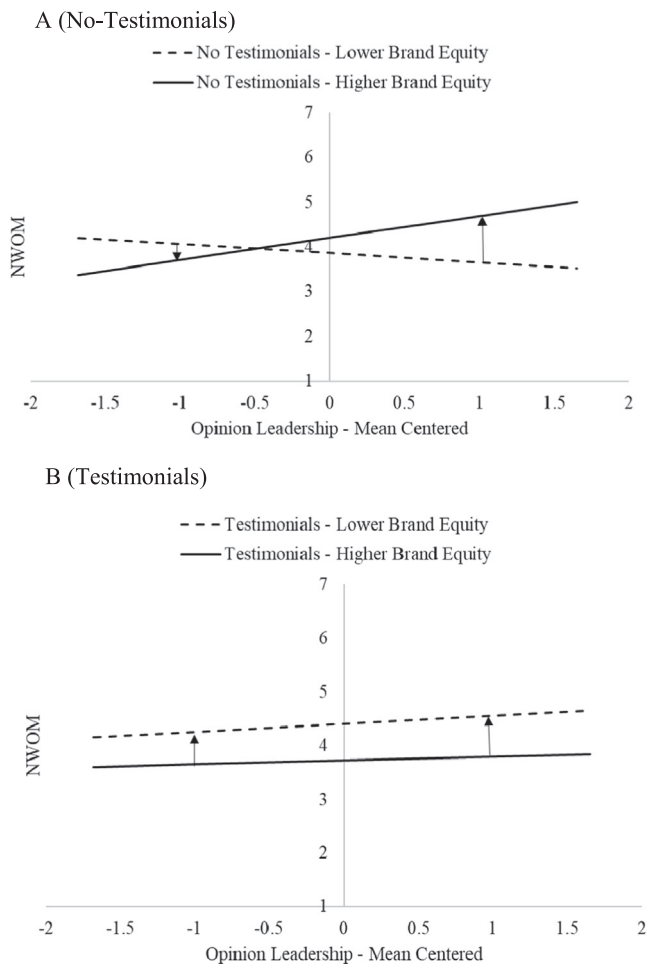

Fig. 2. Study 1: OL × brand equity interactions for NWOM.

above-average OL (higher-OL consumers) and in the presence of testimonials, higher-equity failed brands attract significantly less NWOM compared to lower-equity failed brands (see Fig. 2B). To explicate the interaction, we conduct a spotlight analysis examining the slopes at +1 SD above mean and −1 SD below mean OL. At +1 SD above mean for OL, higher-equity brands attract significantly less NWOM than lower equity brands (*M*'s of 3.81 and 4.6; $\beta = -0.79$, $t = -2.36$, $p = 0.02$). However, there is no such difference at −1 SD below mean OL (*M*'s of 3.63 and 4.21; $\beta = -0.57$, $t = -1.62$, $p = 0.11$).

3.4. Discussion

Study 1 suggests that the vulnerability of higher-equity brands (relative to lower-equity brands) in attracting NWOM following brand failure is conditional on the presence versus absence of accompanying testimonials and to what extent the focal consumers affected by the failure seeks to use the failure to enhance their status as opinion leaders. Higher-equity brand appear to be most vulnerable to attracting NWOM form higher-OL consumers when there are no supporting testimonials to counter the NWOM. Moving ahead to Study 2, we test for a process explanation of our findings.

4. Study 2

Study 2's main objective is to test H2 and H4, i.e., test for a process underlying H1 and H3. In addition, Study 2 aims to enhance the generalizability of Study 1's findings in a different product category (laptops).

4.1. Main study: participants, stimuli and design

Three hundred M-Turk participants (124 females, $M_{Age} = 38.95$ SD = 12.57, all US residents) took part in Study 2 in exchange for monetary compensation. Except for the product category, all aspects of the stimuli of Study 2 were similar to the Study 1.

In Study 2, the participants imagined that they were shopping for a laptop with a longer-lasting battery life (see Appendix A). We manipulated brand equity by using the similar manipulation as in Study 1 (i.e., using fictitious Brand A and with the same dimensions comprising the brand equity construct). The participants imagined that they had narrowed down their search to Brand A (presented as having either higher or lower equity) that was offering a new technology – a two-day battery life without a change in the battery size. They imagined buying the focal brand, and discovering that the battery life was much less than 48 h when they used the laptop for the first time. We manipulated testimonials (absent, present) by giving some participants no further information, and by describing a positive consumer testimonial from a real customer to others.

We randomly assigned the participants to the four conditions obtained by crossing brand equity (higher, lower) and testimonials (present, absent). The sample size, per cell, ranged between 71 and 78 (75 and 76 across higher and lower brand-equity conditions when the testimonial was present, and 78 and 71 when it was absent).

4.2. Measures

All measures are identical to Study 1 except that we replaced facial recognition technology with longer lasting battery life technology where needed (Appendix B). The only new measure introduced is the measure of our mediator, i.e., the opportunity to stand apart by spreading NWOM (three items on 7-point scales, $\alpha = 0.91$). The presentation of items was similar to Study 1 except that we presented the opportunity to stand apart items right after measuring NWOM or the outcome variable.

4.3. Results

4.3.1. Manipulation checks

We conducted the same manipulation checks as in Study 1. The checks targeted brand equity (presented in counterbalanced order), attribute innovativeness/importance, brand/technology failure, focus of blame (self or technology), information load, scenario comprehension and believability, and argument strength.

We conducted an ANOVA on the composite manipulation check ($\alpha = 0.95$), with brand equity (higher, lower), testimonials (present, absent), and order (before/after dependent variable) as predictors. As expected, participants rated Brand A's equity significantly higher in the higher-equity conditions compared to the lower-equity conditions (*M*'s of 5.42 and 3.93, $F(1, 296) = 164.38$, $p < 0.0001$) that was unqualified across the testimonial and order conditions (all interaction *p*'s > 0.05).

4.3.2. Attribute innovativeness and importance

We conducted separate ANOVA's to check how innovative/important participants thought battery life to be, with brand equity and testimonials as predictors. We found no significant main or interaction effects (all *p*'s > 0.05). The innovativeness and importance means for both higher and lower equity brands were significantly above 4.0 or the scale's midpoint (higher equity brand: $M = 4.62$, $t(150) = 5.05$, $p < 0.0001$ and $M = 5.76$, $t(152) = 17.61$, $p < 0.0001$; lower equity brand: $M = 4.56$, $t(145) = 4.75$, $p < 0.0001$ and $M = 5.56$, $t(146) = 15.04$, $p < 0.0001$).

We conducted separate ANOVA's to check how participants rated the performance of battery life, to what extent they thought that the technology had failed and to what extent they blamed themselves or the

device for the failure, with brand equity and testimonials as predictors. We found no significant main or interaction effects (all p's > 0.05). The performance means for the higher and lower-equity brands were all significantly below 4.0 or the scale's mid-point for performance ($M = 3.23$, $t(151) = -4.84$, $p < 0.0001$ and $M = 3.14$, $t(146) = -6.14$, $p < 0.0001$). The failure means for the two brands were all significantly above 4.0 or the scale's mid-point ($M = 5.86$, $t(152) = 16.89$, $p < 0.0001$ and $M = 5.74$, $t(145) = 15.94$, $p < 0.0001$). Finally, the means for blaming the laptop/device were significantly higher than 4.0 or the scale's mid-point for the higher and lower-equity brands, with higher numbers indicating more blame to the device and less blame to the self ($M = 5.95$, $t(152) = 17.34$, $p < 0.0001$ and $M = 5.79$, $t(146) = 15.38$, $p < 0.0001$).

Finally, we compared if adding the testimonial descriptions changes information load, scenario comprehension, scenario believability and argument strength across the different experimental conditions. ANOVAs on these measures with brand equity and testimonials as predictor did not return any significant main or interaction effects (all p's > 0.05).

4.3.3. Hypothesis tests

We test H1 and H3 (moderated moderation model) using the same measures and model as in Study 1 (Model 3; Hayes, 2018). As in Study 1, adding or not adding the covariates and presentation order for the brand-equity manipulation check did not change the statistical significance of the main results and, hence, we report the results for the unadjusted model.

Table 2 shows that the coefficient of equity, conditional on OL = zero (sample average) and testimonial = zero (or absent) is insignificant ($\beta = -0.21$, $t = -1.11$, $p = 0.27$) indicating that, in the absence of testimonials and average OL, both higher and lower-equity failed brands attract the same amount of NWOM. However, the equity by OL interaction is positive and approaches significance ($\beta = 0.26$, $t = 1.77$, $p = 0.08$). The positive sign is consistent with H1 and suggests that, at above-average OL (higher-OL consumers), higher-equity failed brands attract more NWOM than lower equity failed brands (see Fig. 3A). To explicate the interaction, we conduct a spotlight analysis examining the slopes at +1 SD above mean and −1 SD below mean OL. At +1 SD above mean for OL, higher-equity brands attract significantly more NWOM than lower equity brands (M's of 5.65 and 4.61; $\beta = 1.04$, $t = 2.89$, $p < 0.01$) but there is no such difference at −1 SD below mean OL (M's of 4.58 and 5.21; $\beta = -0.64$, $t = -1.61$, $p = 0.11$).

The presence/absence of testimonials further moderates the equity by OL interaction (three-way-interaction $\beta = -0.78$, $t = -2.65$, $p < 0.01$). The negative sign is consistent with H2 and suggests that, at above-average OL (higher-OL consumers) and in the presence of

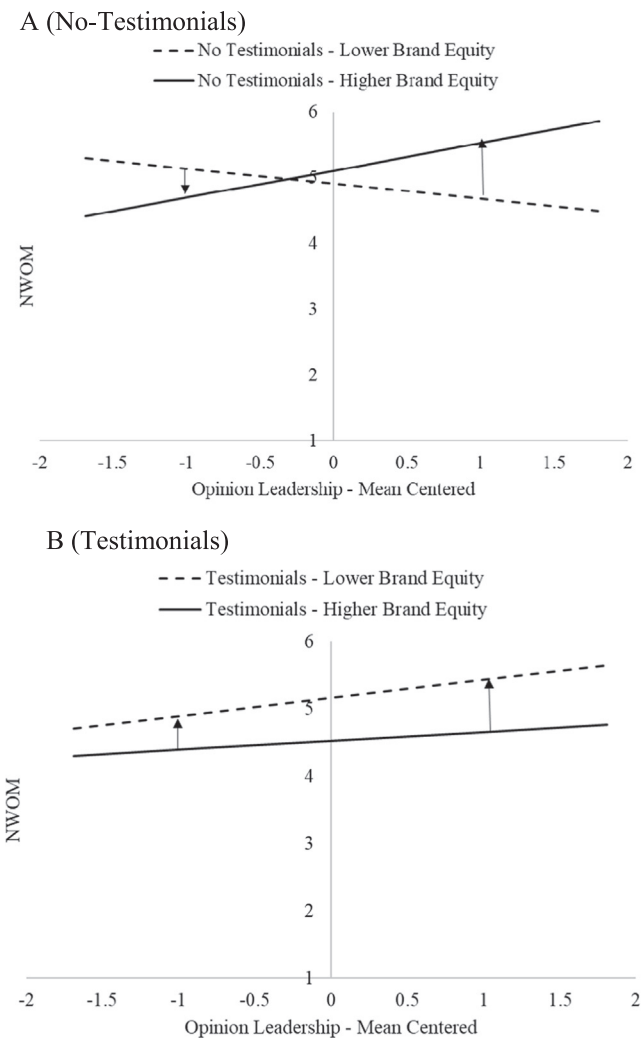

Fig. 3. Study 2: OL × brand equity interactions for NWOM.

Table 2
Study 2: (moderated) moderation model.

A. Dependent Variable: NWOM

	Coefficient	se	t	p	LLCI	ULCI
Constant	4.9301	0.0953	51.7584	0.0000	4.7426	5.1176
Equity (Higher = 1, Lower = 0)	−0.2119	0.1905	−1.1122	0.2670	−0.5870	0.1632
Opinion Leadership (OL)	0.1437	0.0736	1.9515	0.0520	−0.0012	0.2886
Equity × OL	0.2604	0.1473	1.7678	0.0782	−0.0295	0.5503
Testimonial (Present = 1, Absent = 0)	−0.1663	0.1906	−0.8727	0.3836	−0.5414	0.2088
Equity × Testimonial	−0.8454	0.3812	−2.2177	0.0274	−1.5957	−0.0950
OL × Testimonial	0.1103	0.1473	0.7486	0.4548	−0.1797	0.4002
Equity × OL × Testimonial	−0.7818	0.2947	−2.6533	0.0084	−1.3618	−0.2018

B. Conditional effect of brand equity at different values of OL and testimonials

OL	Testimonials	Equity coefficient	se	t	p	LLCI	ULCI
−1 SD Below Mean	Absent	−0.6346	0.3952	−1.6060	0.1094	−1.4125	0.1432
−1 SD Below Mean	Present	−0.4631	0.3601	−1.2860	0.1995	−1.1718	0.2457
+1 SD Above Mean	Absent	1.0387	0.3597	2.8876	0.0042	0.3306	1.7468
+1 SD Above Mean	Present	−0.8236	0.4114	−2.0022	0.0462	−1.6333	−0.0139

NWOM = Negative Word of Mouth, OL = Opinion Leadership, se = Standard Error, LLCI = Lower-Level Confidence Interval, ULCI = Upper-Level Confidence Interval.

Table 3
Study 2: process test.

A. Dependent variable: opportunity to stand apart

	Coefficient	se	t	p	LLCI	ULCI
Constant	3.8841	0.0851	45.6447	0.0000	3.7166	4.0516
Equity (Higher = 1, Lower = 0)	−0.2183	0.1702	−1.2823	0.2008	−0.5533	0.1168
Opinion Leadership (OL)	0.5415	0.0658	8.2321	0.0000	0.4120	0.6710
Equity × OL	0.2281	0.1316	1.7336	0.0841	−0.0309	0.4871
Testimonial (Present = 1, Absent = 0)	0.0734	0.1702	0.4314	0.6665	−0.2617	0.4085
Equity × Testimonial	−0.8033	0.3405	−2.3589	0.0190	−1.4736	−0.1330
OL × Testimonial	−0.0726	0.1316	−0.5518	0.5815	−0.3316	0.1864
Equity × OL × Testimonial	−0.6700	0.2632	−2.5452	0.0115	−1.1881	−0.1518

B. Dependent variable: intent to spread word-of-mouth

	Coefficient	se	t	p	LLCI	ULCI
Constant	4.1316	0.2476	16.6849	0.0000	3.6442	4.6190
Brand (High Equity = 1, Low Equity = 0)	−0.1241	0.1898	−0.6537	0.5139	−0.4977	0.2496
Opportunity to Stand Apart (OSA)	0.2121	0.0583	3.6406	0.0003	0.0974	0.3268

C. Conditional indirect effect of brand on NWOM via OSA at different values of moderators

OL	Testimonial	Indirect effect	Bootstrapped se	Bootstrapped LLCI	Bootstrapped ULCI
−1 SD Below Mean	Absent	−0.1163	0.0922	−0.3650	0.0140
−1 SD Below Mean	Present	−0.1019	0.0785	−0.3054	0.0155
+1 SD Above Mean	Absent	0.1906	0.0961	0.0524	0.4489
+1 SD Above Mean	Present	−0.1646	0.0990	−0.4365	−0.0244

NWOM = Negative Word of Mouth, OL = Opinion Leadership, se = Standard Error, LLCI = Lower-Level Confidence Interval, ULCI = Upper-Level Confidence Interval.

testimonials, higher-equity failed brands attract significantly less NWOM compared to lower-equity failed brands (see Fig. 3B). To explicate the interaction, we conduct a spotlight analysis examining the slopes at +1 SD above mean and −1 SD below mean OL. At +1 SD above mean for OL, higher-equity brands attract significantly less NWOM than lower equity brands (*M*'s of 4.69 and 5.51; $\beta = -0.82$, $t = -2.00$, $p = 0.05$) but there is no such difference at −1 SD below mean OL (*M*'s of 4.32 and 4.53; $\beta = -0.46$, $t = -1.29$, $p = 0.20$).

4.3.4. Testing for the mediating role of opportunity to stand apart

To test our mediation model (Fig. 1B), we used Hayes' Model 11 (Hayes, 2018). The mediator is the opportunity to stand apart (three-item composite scale). The predictor (brand equity), the outcome (NWOM) and the moderators (OL and consumer testimonials) are the same as before. As before, adding or not adding the covariates and presentation order for the brand-equity manipulation checks did not change the statistical significance of the main results and, hence, we report the results for the unadjusted model. Table 3 shows the results for the unadjusted model.

Table 3's top panel tests how OL and testimonials moderate the predictor (brand equity) to mediator (opportunity to stand apart) path. The coefficient of equity, conditional on OL = zero (sample average) and testimonial = zero (or absent) is insignificant ($\beta = -0.22$ $t = -1.28$, $p = 0.20$). Thus, in the absence of testimonials and average OL, spreading NWOM about the failures of higher or lower-equity brands do not provide any differential opportunities to stand apart. However, the equity by OL interaction is positive and approaches significance ($\beta = 0.23$, $t = 1.73$, $p = 0.08$). The positive sign is consistent with H2. Spreading NWOM about higher-equity failed brands provides more opportunity to stand apart compared to spreading NWOM about lower-equity brand failures at above-average OL (among higher-OL consumers; see Fig. 4A). To explicate the interaction, we conduct a spotlight analysis examining the slopes at +1 SD above mean and −1 SD below mean OL. At +1 SD above mean for OL, spreading NWOM about failed higher-equity brands afford more opportunity to stand apart compared to spreading NWOM about failed lower equity brands (*M*'s of 5.05 and 4.15; $\beta = 0.90$, $t = 2.80$, $p < 0.01$). However, at −1SD below mean this difference is negative but not significant (*M*'s of 2.82 and 3.37; $\beta = -0.55$, $t = -1.55$, $p = 0.12$).

The presence/absence of testimonials further moderates the equity by OL interaction (three-way-interaction $\beta = -0.67$, $t = -2.55$, $p = 0.01$). The negative sign is consistent with H4. At above-average OL level (higher-OL consumers) and in the presence of testimonials, spreading NWOM about higher-equity failed brands provides less opportunity to stand apart compared to spreading NWOM about lower-equity failed brands (see Fig. 4B). To explicate the interaction, we conduct a spotlight analysis examining the slopes at +1 SD above mean and −1 SD below mean OL. At +1 SD above mean for OL, spreading NWOM about failed higher-equity brands afford less opportunity to stand apart compared to spreading NWOM about failed lower equity brands (*M*'s of 4.19 and 4.96; $\beta = -0.78$, $t = -2.11$, $p = 0.04$). However, at −1SD below mean, this difference is not significant (*M*'s of 3.02 and 3.50; $\beta = -0.48$, $t = -1.49$, $p = 0.14$).

As shown in Table 3's middle panel, brand equity does not affect the NWOM once we factor in the mediator (the opportunity to stand apart). However, the opportunity to stand apart positively affects NWOM, even after controlling for brand equity ($\beta = 0.21$, $t = 3.64$, $p < 0.001$) and indicates that there is a mediation path as depicted in Fig. 1B.

Finally, as shown in Table 3's bottom panel, the mediation is conditional on both OL and the presence/absence of testimonials. Using two spotlights on the OL-scale, at 1-SD above the mean and another at 1-SD below the mean, we find that the indirect paths are significant only at 1-SD above the mean and reverse sign across the two testimonial conditions.

In the absence of testimonials, the indirect path is positive and the 95% bootstrapped confidence interval is fully positive (0.05, 0.45). It indicates that spreading NWOM about failed higher-equity brands allows the higher-OL consumer to stand apart more compared to spreading NWOM about failed lower-equity brands (a positive effect). Furthermore, the opportunity to stand apart increases her/his tendency to spread NWOM (another positive effect).

In the presence of testimonials, the indirect path is negative and the

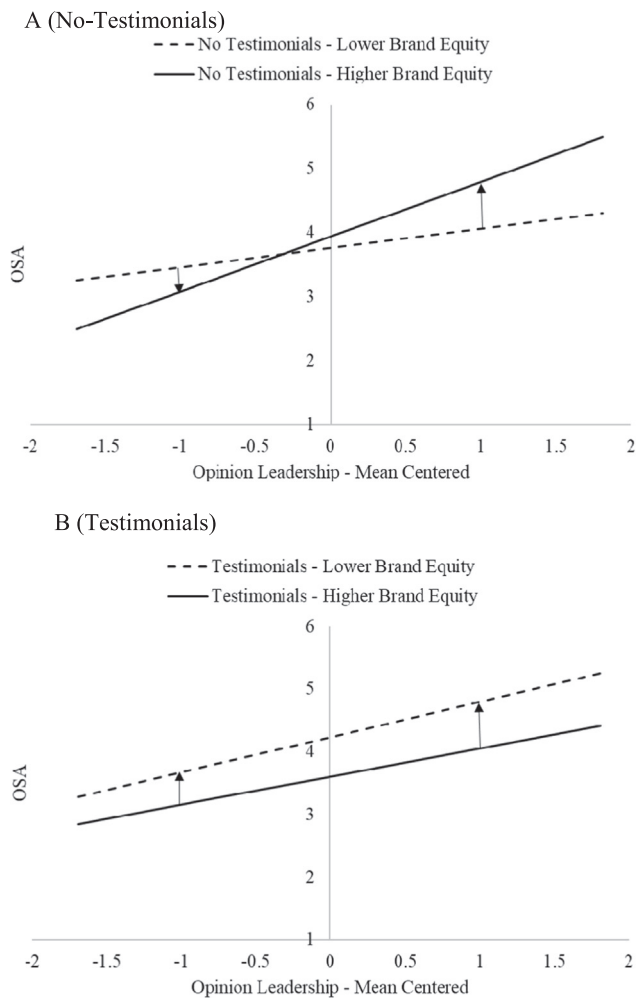

Fig. 4. Study 2: OL × brand equity interactions for opportunity to stand apart (OSA).

95% bootstrapped confidence interval is fully negative (–0.43, –0.02). It indicates that spreading NWOM about failed higher-equity brands provides less opportunity to the higher-OL consumer to stand apart compared to spreading NWOM about failed lower-equity brands (a negative effect). Furthermore, the lower opportunity to stand apart reduces her/his tendency to spread NWOM (a positive effect).

4.4. Discussion

Study 2 makes two contributions. First, we replicate Study 1's results in a different product class. More importantly and second, we test for a process explanation of the findings by testing a moderated mediation model describing a three-part process. First, we show that, whether the failure of higher/lower equity brands encourage/discourage attracting NWOM from a consumer will depend upon the extent such NWOM enables the consumer's opinion to stand apart from others (a mediation process). Moreover and second, we show that OL (a consumer characteristic) and the presence/absence of testimonials (a brand characteristic) moderate this indirect path. More specifically, among higher-OL consumers, failed higher-equity brands attract more NWOM than failed lower-equity brands in the absence of testimonials. However, among the same higher-OL consumers, failed higher-equity brands attract less NWOM than failed lower-equity brands in the presence of testimonials.

5. General discussion

NWOM from opinion leaders has the power to slow down the adoption process (Goldenberg et al., 2009). In a marketplace where almost half of the innovations fail, it is, therefore, important for marketers to understand what motivates these opinion leaders to spread NWOM (their motivation) about a target (focal) brand and what brand managers can do to thwart their negative messages. Our research indicates that, when the marketer takes no pre-emptive measures in the form of consumer testimonials, these consumers will target failed higher-equity brands more than failed lower-equity brands to spread NWOM. However, when marketers have supportive consumer testimonials for their innovations, they will target lower-equity brands more than higher-equity brands to spread NWOM.

5.1. Theoretical contributions

The relationship between branding and innovations is a growing, but still a nascent area in marketing (Brexendorf et al., 2015). In the case of an innovation failure, the current literature argues that there are both advantages and disadvantages of being a higher-equity brand. For example, whereas Choi and Mattila (2008) show that higher equity often shields the brand when a failure occurs, Niedrich, Kiryanova, and Black (2005) show that the higher equity exacerbates frustration and disappointment. Liao and Cheng (2014) try to resolve these competing views by showing that (1) higher-equity brands are better able to weather failures as long as there are no preannouncements to create higher expectations, and (2) positive word of mouth support is more critical in weathering failure for lower-equity brands compared to higher-equity brands.

Our work fits nicely with this emerging research in the area of brand equity and new product introductions. We find that consumer testimonials protect higher-equity brands to the extent that they discourage higher-OL consumers (those at above-average OL) from spreading NWOM about their failure. These testimonials, however, do not protect lower-equity brands. In fact, the presence of these testimonials encourages the same consumers to spread NWOM about their failure, something that they are unlikely to do when there are no supporting testimonials accompanying these brands.

Although scholars have investigated why consumers choose to spread WOM (Zhang et al., 2014), there is less research on the relationship between WOM and branding (Lovett et al., 2013). We extend the literature on WOM by investigating how the interaction of opinion leadership, brand equity, and consumer testimonials lead to the generation of NWOM. The inclusion of opinion leadership as a construct of interest is theoretically valid since opinion leadership is an important vehicle for diffusing the innovation information (Ruvio & Shoham, 2007).

Our research also contributes to the literature on communication strategies in general and consumer testimonials particular. The main objective of consumer testimonials is to use a credible source (an actual user) to persuade consumers (Martin et al., 2008). Ordinarily, this should help a lower-equity brand more than a higher-equity brand. However, our results show that adding a positive content to a weak (lower-equity) brand compared to a strong (higher-equity) brand may, unintentionally, attract a negative market reaction and encourage some consumers to spread NWOM if the brand fails.

5.2. Managerial implications

Our research has important managerial implications. First, the findings confirm the conventional wisdom that brand managers need to identify and communicate differentially with consumers who desire to be opinion leaders, or who are recognized in the market as opinion leaders for a specific product category; e.g., Chaney, 2001; Shi & Wojnicki, 2014). These higher-OL consumers, compared to the average,

are more involved, knowledgeable, and accepted as a source of information by their peers. In return, they are looking for opportunities to reinforce their status. Brand managers, especially of higher equity brands, could prioritize these consumers for after sales services to understand and fix potential product related problems sooner than later since these consumers will more likely target their brands to spread NWOM. Similarly, brand managers could invest in building closer relationships with these consumers by sending them more information about the innovation prior to, or during, the product launch. Finally, brand managers should be careful when designing consumer testimonials. While research indicates the effectiveness of this marketing communication tool (e.g., Martin et al., 2008), our findings point to a more nuanced understanding. More specifically, brand managers are better off pairing consumer testimonials with higher-equity brands (accentuating their advantage), but not when pairing testimonials with lower-equity brands (reducing their disadvantage).

5.3. Limitations and future research

We conclude by addressing a few limitations of our studies and offer directions for future research. First, we use self-reports of opinion leadership, which, research indicates, is more influential within the network of strong ties (Risselada, Verhoef, & Bijmolt, 2016). This would indicate that we should also consider to whom the opinion leaders would talk and to what extent the intended audience would react to arrest the diffusion of the innovation. Hence, future research could investigate the impact of strong versus weak social ties on the tendency to spread NWOM. For example, do higher-OL consumers tend to spread NWOM more when they have strong ties within their network?

Second, future research could investigate if and how the contents of the actual NWOM change depending on the context. For example, text-based analytics can quantify the emotional content of the NWOM and determine if the emotional tone is stronger when the focal brand is a higher-equity brand and muted when the focal brand is a lower-equity brand.

Third, the mediator in our process explanation is the consumer's desire to stand apart. As research indicates, quite often the risk of being wrong acts as a check to this desire (Schlosser, 2009). Future research can investigate the interplay between the two, in the same way as we do for risk/return trade-offs. Consumer testimonials may enhance the salience of the risk dimension and thereby deter the opinion leaders from spreading NWOM about higher-equity brands. There needs to be research to test this proposition.

Fourth, we do not tap into the behaviour of the lower-OL consumers. Although researchers often categorize them as passive followers, it would be interesting to understand what drives their indifference to brand equity when spreading NWOM about failed brands. It could be that they are not involved with the brands and thus do not care if it is a higher-equity or lower-equity brand that has failed. Alternatively, it could be their lack of knowledge differentiating between higher and lower-equity brands. Future research could screen participants on their level of opinion leadership and then assign them randomly to different experimental conditions. Another interesting area would be to identify if there is a change in motivation to spread NWOM across different product categories by the same consumer. For example, a consumer may have very little desire to be an opinion leader in one category but have a strong desire to be so in another category.

Finally, we focused on product failures in our two studies – for example, the facial recognition payment failed in Study 1 and the laptop battery failed in Study 2. Consumers may have construed the scenarios as a mixture of product and service failure. For example, if consumers discover that the facial recognition device/laptop battery is not living up to the promised standards, they might think that this is an installation (service) fault rather than a product fault. We know that consumers differ in how they react to product and service failures. For example, recent research indicates that when consumers experience a service failure (e.g., a delayed flight), how they emotionally cope with the failure determines what they do about the problem (coping mechanism; see Sengupta, Balaji, & Krishnan, 2015). Such coping mechanisms include positive thinking (looking at the brighter side), avoidance (not thinking of the experience), and emotional support (relying on others to feel better). Research has shown that, for less severe failures (flight delayed by two hours), higher-equity brands foster significantly greater positive coping compared to lower-equity brands (Sengupta et al., 2015). Matching the desire to spread NWOM with coping mechanisms across higher and lower-equity brands constitute an interesting avenue for future research.

Appendix A. Study scenarios

Study 1	Study 2
Imagine you are looking to choose a new bank because your current bank is not convenient for your credit card payments. You do your research and talk to your friends about their experiences with different bank brands.	Imagine you are looking to buy a new laptop that has better battery life than your current laptop. You do your research and talk to your friends about their experiences with laptop brands.
You narrow down your search to Brand A. Your attitude towards Brand A is very positive. Your view of Brand A's image is very positive and feel that the perceived quality of Brand A is very high. You think that your future loyalty towards Brand A will be very high and you are willing to pay a higher price for Brand A's services compared to the average service price of all banks in the market.	You narrow down your search to Brand A. Your attitude towards Brand A is very positive. Your view of Brand A's image is very positive and feel that the perceived quality of Brand A is very high. You think that your future loyalty towards Brand A will be very high and you are willing to pay a higher price for Brand A compared to the average price of all laptop brands sold in the market.
(Your attitude towards Brand A is about average. Your view of Brand A's image is about average and feel that the perceived quality of Brand A is about average. You think that your future loyalty towards Brand A will be about average and you are willing to pay the average market price for Bank A's services, i.e., *the average service price of all banks in the market.)*	*(Your attitude towards Brand A is about average. Your view of Brand A's image is about average and feel that the perceived quality of Brand A is about average. You think that your future loyalty towards Brand A will be about average and you are willing to pay the average market price for Brand A,* i.e., *the average price of all laptop brands sold in the market.)*
Brand A offers a new technology in the banking market. This is a facial recognition device. These devices are readily available at the retail outlets you visit. It uses an algorithm that recognizes you and debits your card if you nod at the screen for approval. Brand A promises that using this device will offer great convenience as you will not need to carry your card or even your phone.	Brand A offers a new battery technology in the laptop market. This new battery technology allows you to use your computer for 48 h without charging. Therefore, your battery will be same in size but have a two-day life.
You decide to choose Brand A as your bank and start using the facial recognition payment at different retailers. The first time you try facial recognition, you realize that it <u>does not recognize</u> you and you will have to pay using the credit card. This happens despite you and the retail person following all the instructions the bank provided.	You decide to buy the laptop from Brand A. The first time you use your laptop, you realize the battery lasts <u>a lot less</u> than Brand A advertised and it is only at the level of a regular laptop. This happens despite you following all the instructions in the user manual.
<u>Testimonial Condition:</u>	*<u>Testimonial Condition:</u>*
Later in the day, you come across an ad from Brand A. The ad features a testimonial from a real customer, Mr. Robert Brown, who says,	Later in the day, you come across an ad from Brand A. The ad features a testimonial from a real customer, Mr. Robert Brown, who says,

"I switched to Brand A bank to benefit from the new facial recognition system they offer. This is great! I don't need to carry anything for my payments!"	*"I bought the new Brand A laptop to benefit from the 2-day battery life they offer. This is great! I don't need to charge my laptop for 2 days!"*

Appendix B. Scale items – Listed alphabetically

Construct	Scale items (All 7-point scales)	Note
Attribute Innovativeness (Bouncken, Ratzmann, & Winkler, 2008; Study 1 and Study 2)	*Strongly disagree / strongly agree* scales (1) The technology this product incorporates is new to you (2) The benefits this product offers are new to you (3) You perceive the product features as novel/unique (4) This product introduces many completely new features to the market	In Study 1, a factor analysis using Maximum Likelihood method with Varimax rotation on the four items retained a one-factor solution (eigenvalue > 1) with factor loadings between 0.76 and 0.89. In Study 2, a factor analysis using Maximum Likelihood method with Varimax rotation on the four items retained a one-factor solution (eigenvalue > 1) with factor loadings between 0.71 and 0.94.
Argument Believability (Study 1 and Study 2)	(1) I think that the scenario is (*not believable / believable*) (2) I think that the scenario is (*not credible / credible*)	
Argument Comprehension (Study 1 and Study 2)	(1) I think that the scenarios is (7-point, *difficult to understand / easy to understand*)	
Argument Strength (Study 1 and Study 2)	(1) I think that the scenario is (7-point, *not convincing / convincing*) (2) I think the arguments in the scenario are (7-point, *weak / strong*)	
Brand Equity (Study 1 and Study 2)	(1) What kind of attitude do you have about Brand A ___ (*negative / positive*) (2) What kind of image does Brand A ___ have (*negative / positive*) (3) How would you rate the quality delivered by Brand A ___ (*low / high*) (4) How loyal are you to Brand A ___ (*not at all / very*) (5) Would you be willing to pay more for Brand A ___ than you would another ___ (*definitely not / definitely*)	Note: In Study 1, a factor analysis using Maximum Likelihood method with Varimax rotation on the five items retained a one-factor solution (eigenvalue > 1) with factor loadings between 0.79 and 0.92. In Study 2, a factor analysis using Maximum Likelihood method with Varimax rotation on the five items retained a one-factor solution (eigenvalue > 1) with factor loadings between 0.82 and 0.92.
Brand Performance (Study 1 and Study 2)	I think the performance of ____'s new technology is: (*very bad / very good*)	
Failure Attributions (Study 1 and Study 2)	(1) My bad experience with the brand is ___ *attributed completely to me / attributed completely to Brand A*) (2) I have had this bad experience because the new technology by Brand A failed (*strongly disagree / strongly agree*)	
Information Load (Burton, Biswas, & Netemeyer, 1994; Aljukhadar, Senecal, & Daoust, 2010; Study 1 and Study 2)	(1) The scenario I read has (*not enough information / too much information*) (2) I require more information before I can evaluate Brand A's performance (*strongly disagree / strongly agree* - reverse coded)	
Negative word-of mouth intentions (Price & Arnould, 1999; Study 1 and Study 2)	*Strongly disagree / strongly agree* scales (1) I will recommend that they not choose Brand A to anyone who seeks my advice (2) I will say negative things about Brand A to other people (3) I will discourage others to buy Brand A	In Study 1, a factor analysis using Maximum Likelihood method with Varimax rotation on the three items retained a one-factor solution (eigenvalue > 1) with factor loadings between 0.69 and 0.91. In Study 2, a factor analysis using Maximum Likelihood method with Varimax rotation on the three items retained a one-factor solution (eigenvalue > 1) with factor loadings between 0.71 and 0.94.
Opinion Leadership (Childers, 1986; Study 1 and Study 2)	(1) In general, do you talk to your friends and neighbors about (product) (*never / very often*)? (2) When you talk to your friends and neighbors about (product) do you (*give very little information / give a great deal of information*) (3) During the past six months, how many people have you talked to about (product) (*no one / a number of people*) (4) Compared to your circle of friends, how likely are others to ask you about (product) (*not at all likely / very likely to be asked*) (5) In discussing (product), which of the following happens most often: (*your friends tell you about (product) / you tell your friends about (product)*) (6) Overall, in all of your discussions with friends and neighbors are you: (*not used as source of advice / often used as a source of advice*)?	Note: In Study 1, a factor analysis using Maximum Likelihood method with Varimax rotation on the six items retained a one-factor solution (eigenvalue > 1) with factor loadings between 0.59 and 0.86. In Study 2, a factor analysis using Maximum Likelihood with Varimax rotation on the six items retained a one-factor solution (eigenvalue > 1) with factor loadings between 0.65 and 0.80.
Opportunity to Stand Apart (Study 2)	*Strongly disagree / strongly agree* scales (1) Letting others know of my experience with Brand A will allow me to show my knowledge on laptops (2) Letting others know of my experience with Brand A is a chance to stand out as an expert among other people (3) Given this is Brand A, people would think highly of me if I let them know of my experience with it	A factor analysis using Maximum Likelihood method with Varimax rotation on the three items retained a one-factor solution (eigenvalue > 1) with factor loadings between 0.79 and 0.86.
Satisfaction (Oliver, 1980; Study 1 and Study 2)	(1) Based on your experience with Brand A, how satisfied/dissatisfied are you (*extremely dissatisfied / extremely satisfied*) (2) Based on your experience with Brand A, how contented/frustrated are you (*frustrated/contended*)	

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